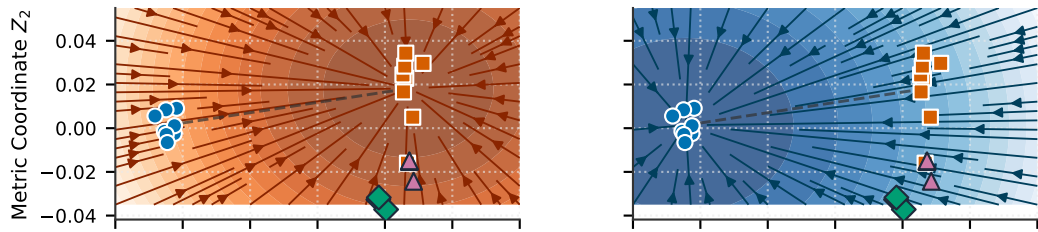


A Metric Force Field & Hookean Restoring Potential (HIV-1 pol)

Protease Flank (0-862 nt)
Basin at Subtype C (Home)

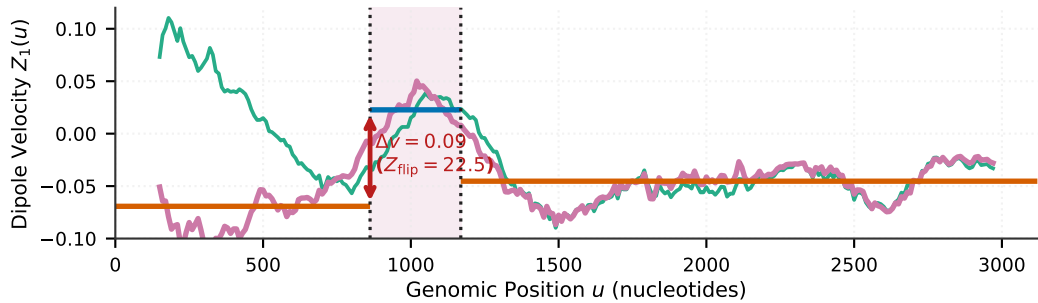
RT Cassette (862-1169 nt)
Basin at Subtype B (Donor)

● Subtype B (Donor) ● Subtype C (Home) ◆ CRF07_BC ($n = 3$) ▲ CRF08_BC ($n = 3$)



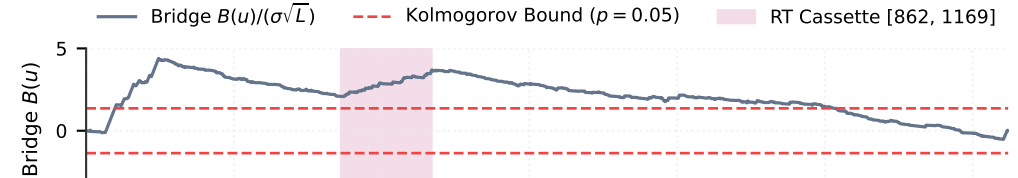
Dynamic Velocity Reversal Along Subtype B/C Dipole Axis ($Z_{\text{flip}} = 22.5$, $p < 10^{-15}$)

— CRF07_BC (AF286230) — CRF08_BC (AY008715) — PR Flank ($\bar{Z}_1 = -0.07$) — RT Cassette ($\bar{Z}_1 = 0.02$)

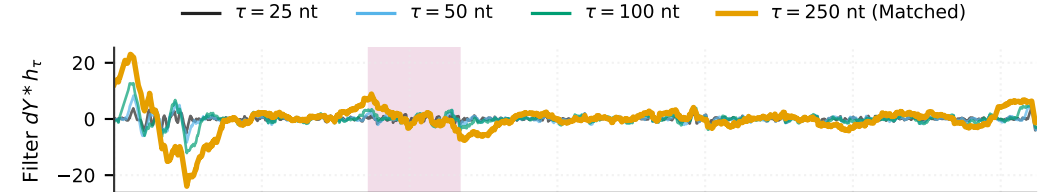


B Multi-Scale Dyadic Filters & Running Drawdown (HIV-1 pol)

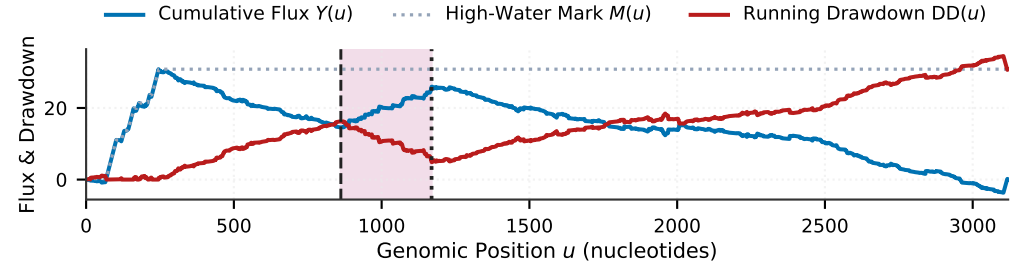
1D Brownian Bridge Dilution (Global Drift Peaks at $u = 2450$; Cassette Diluted)



Multi-Scale Dyadic Wavelets: Matched Scale $\tau = 250$ nt Resonates at Cassette



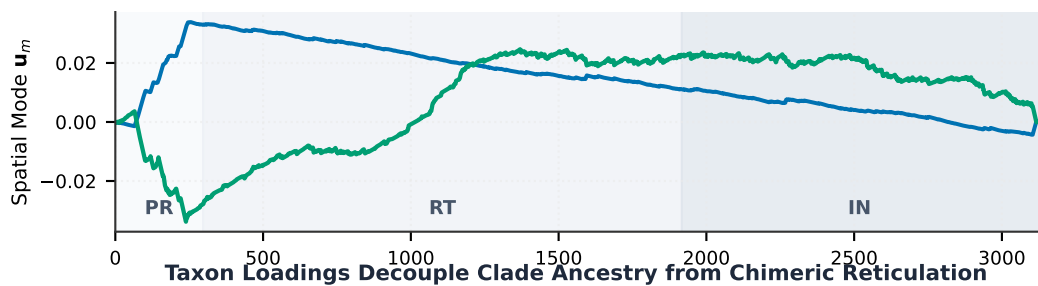
Running Drawdown Localizes Cassette Entry & Momentum Collapse at Exit ($u_2 = 1169$)



C Spatio-Temporal Force Field & SVD Mode Peeling (HIV-1 pol)

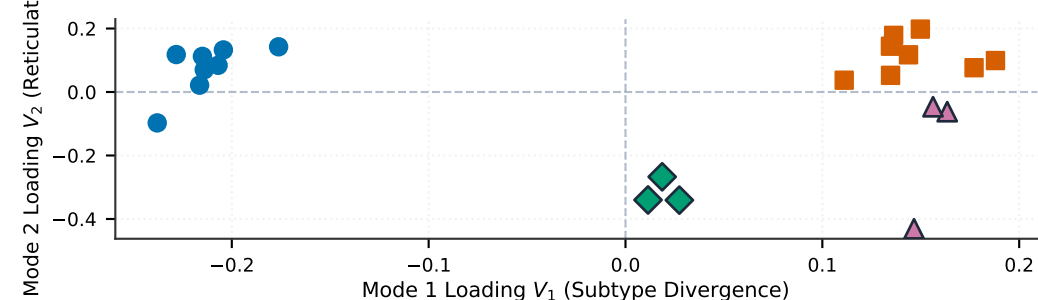
Spatial Profiles: Mode 1 (Clade Backbone) vs Mode 2 (CRF07/08 Mosaicism)

— Mode 1 (Clade Backbone) — Mode 2 (CRF07/08 Mosaicism)



Taxon Loadings Decouple Clade Ancestry from Chimeric Reticulation

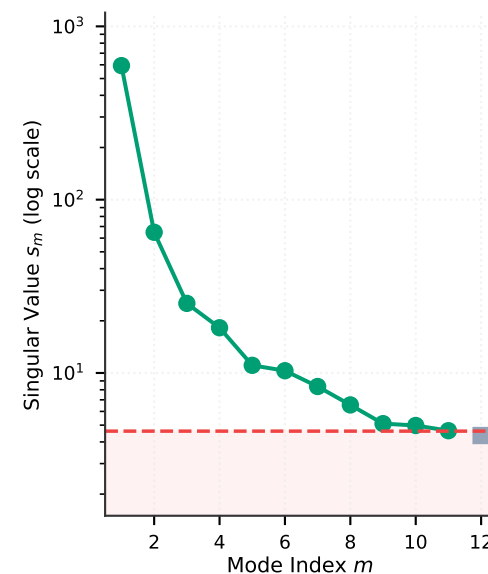
● Subtype B ($n = 8$) ● Subtype C ($n = 8$) ◆ CRF07_BC ($n = 3$) ▲ CRF08_BC ($n = 3$)



D Random Matrix Theory Halting & Continuous Topological Mode Ratio

RMT Spectral Halting Edge
($S_{\text{bulk}} = 4.6$; Halts at Mode 12)

● Signal ($s_m > 4.6$) ■ Noise — Edge (4.6)



Topological Mode Classification
(Monopolar Backbone vs Bipolar Cassette)

— Mode 1 Backbone ($p = 0.97$) — Mode 2 Cassette ($p = 0.48$)

